

Matrices: Addition and Scalar multiplication

Definition

- 1 Suppose that A, B are $m \times n$ matrices then we define the product $A + B$ to be the $m \times n$ matrix with ij^{th} entries $a_{ij} + b_{ij}$.
- 2 Suppose A is an $m \times n$ matrix and λ is a scalar then λA is the $m \times n$ matrix with ij^{th} entries λa_{ij} .

Ex:

$$i) \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \\ 2 & 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 2 & -1 \\ 2 & 4 & 5 \end{pmatrix}$$

$$ii) \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \times \text{ No!}$$

$$iii) 2 \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 6 & 8 \end{pmatrix}$$

If A, B have the same size then

$$A + B = B + A$$

If λ, μ are scalars then

$$\lambda(A + B) = \lambda A + \lambda B.$$

$$\lambda(\mu A) = (\lambda\mu)A.$$

Matrices: Multiplication

Definition

Suppose that A is an $m \times k$ matrix and $B = [b_1 \ b_2 \ \dots \ b_n]$ is a $k \times n$ matrix then we define the product AB by

$$[Ab_1 \ Ab_2 \ \dots \ Ab_n]$$

which is an $m \times n$ matrix.

Matrix multiplication as composition

If A is $m \times k$ and B is $k \times n$ then AB is the standard matrix of the composition of the matrix transformations $T_A \circ T_B$.



Ex:

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \leftarrow B$$

2×3 A 3×2

$$= \begin{pmatrix} 5 & 6 \\ -2 & -1 \end{pmatrix}$$

2×2

$$\begin{pmatrix} 1 & 2 \\ 0 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & -2 \end{pmatrix}$$

$\nearrow$
 3×2
 $\nwarrow$
 2×3

$$= \begin{pmatrix} 2 & -1 & -1 \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{pmatrix}$$

$\nearrow$
 3×3

$$AB \neq BA.$$

$\nwarrow$ $\nearrow$
 not the same
 size.

Ex: What if A and B were 2×2 ?
 $\uparrow$
 i.e. square.

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

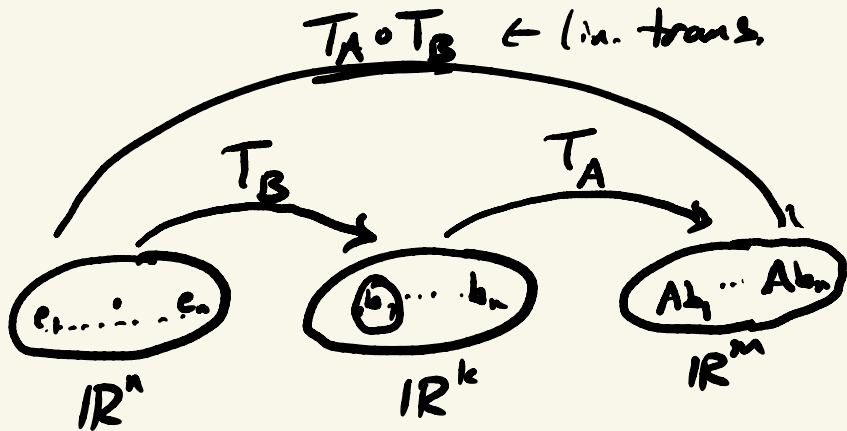
$$AB = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 6 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 1 \end{pmatrix}$$

$AB \neq BA$. Matrix mult is not commutative.

$$\begin{array}{l} A \rightarrow m \times k. \\ \underline{B \rightarrow k \times n.} \end{array}$$

$$AB \rightarrow m \times n.$$



$$T_B(e_i) = b_i.$$

$$B = (b_1 \dots b_k)$$

The standard matrix for

$$T_A \circ T_B = (Ab_1, \dots, Ab_n) \\ = AB.$$

Matrix Algebra

Definition

The $n \times n$ matrix with diagonal entries 1 and all other entries 0 is called the $n \times n$ identity matrix and is denoted I_n .

For all matrices A, B and C of the correct size and scalars λ

- $A(BC) = (AB)C$ associative.
 - $A(B + C) = AB + AC$
 - $(A + B)C = AC + BC$
- } distributivity.
- $\lambda(AB) = (\lambda A)B = A(\lambda B)$
 - If A is $m \times n$ then $I_m A = A = A I_n$

Ex:

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

$n \times n$
identity
matrix.

$\hat{I}_{n \times n}$

1's on the
diagonal
0 everywhere
else.

$$\underline{I_n A = A = A I_n}, \quad A \text{ is } n \times n.$$

Ex:

$$\underline{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}} \underline{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

↑
0 matrix 2×2 .

Two non-zero matrices
mult. together to give 0.

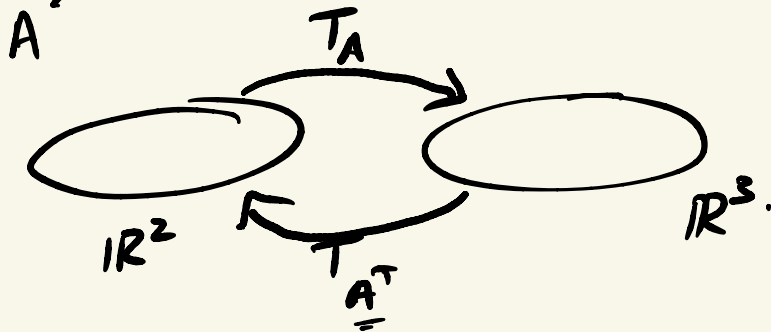
Matrices: Powers and the transpose

- If A is an $n \times n$ matrix then for an k we can multiply A together k times and write A^k . We say $A^0 = I_n$. 
- If $A = (a_{ij})$ is $m \times n$ then the transpose of A , written A^T , is the $n \times m$ matrix where the ij entry is a_{ji} i.e. the rows of A become the columns of A^T and vice versa.

For all matrices A and B of the correct size and scalars λ

- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$
- $(\lambda A)^T = \lambda A^T$

$$E_r \quad A \rightarrow \begin{pmatrix} 2 & 3 \\ 1 & 0 \\ -1 & 2 \end{pmatrix}^T = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 0 & 2 \end{pmatrix}$$



$$AA^T$$

3×3

or $A^T A$

2×2 .

Definition

We say an $n \times n$ matrix A is invertible if there is a matrix B such that $AB = BA = I_n$. We call B the inverse of A and write A^{-1} for B .

Fact:

- If A is invertible then A^{-1} is uniquely defined.
- If A is invertible then the associated linear transformation is one-to-one and onto.

A little bit of algebra with inverses

- If A is invertible then so is A^{-1} and $(A^{-1})^{-1} = A$.
- If A and B are invertible and of the same size then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is $n \times n$ and invertible then so is A^T and $(A^T)^{-1} = (A^{-1})^T$.